NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE NAME

CLASS

BIOLOGY

9648/01

Paper 1 Multiple Choice

September 2015

1 hour 15 min

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name and CT on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
Calculators may be used.

This document consists of **20** printed pages and **0** blank page.

[Turn over

1 Membranous sacs containing products of metabolism are formed by the endoplasmic reticulum in cells.

Where are these products used?

A inside and outside the cell
B inside lysosomes only
C inside the cell only
D outside the cell only

2 Which correctly matches the functional and structural features of cellulose, collagen, glycogen and triglyceride?

		function	structure		
			fibrous	molecule held together by hydrogen bonds	branched chains
A	cellulose collagen	support strengthening	✓ ✓	✗ ✓	✓ ✗
B	cellulose triglyceride	support energy source	✓ ✗	✓ ✗	✗ ✗
C	collagen glycogen	strengthening storage	✓ ✗	✓ ✗	✓ ✓
D	glycogen triglyceride	storage energy source	✗ ✗	✓ ✓	✓ ✗

key ✓ = true ✗ = false

3 Ethylene glycol is a chemical used to prevent water from freezing. If ethylene glycol is swallowed accidentally, it is metabolised by an enzyme found in liver cells to produce a toxic product.

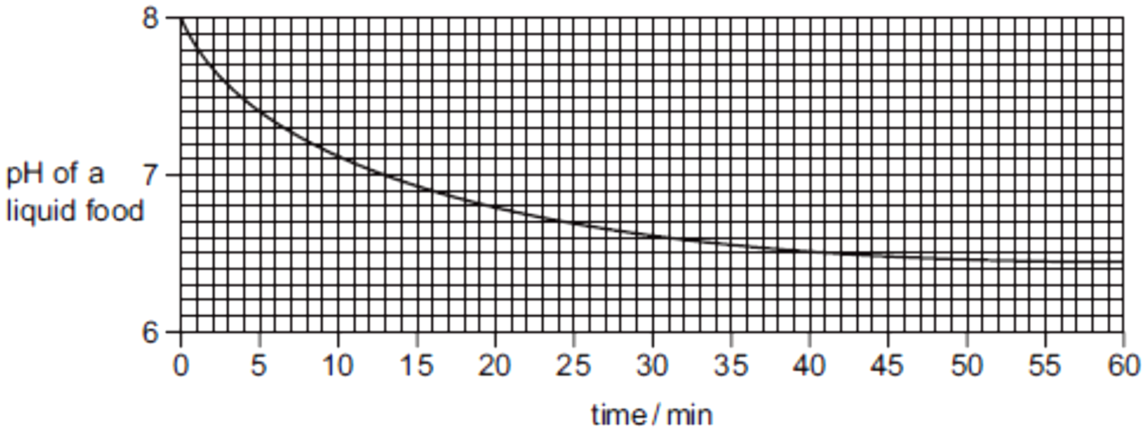
The enzyme normally catalyses the oxidation of ethanol to a harmless product.

People who have swallowed ethylene glycol are treated with large doses of ethanol. This prevents formation of a toxic product and allows the body to excrete the ethylene glycol.

Which statement describes why this treatment works?

A Ethanol binds near the active site on the enzyme, altering its shape.
B Ethanol binds permanently to the active site of the enzyme, blocking it.
C Ethanol changes the tertiary structure of the enzyme, denaturing it.
D Ethanol is more likely to bind to the active site on the enzyme.

4 Lipase is a digestive enzyme produced by the pancreas that catalyses the hydrolysis of dietary lipids. The table shows how the pH of a liquid food containing a high proportion of lipids decreases over time.



Which statements are possible explanations of the results of the experiment between 50 and 60 minutes?

1 Enzyme concentration becomes the limiting factor.
2 Substrate concentration becomes the limiting factor.
3 All the enzyme active sites are saturated.
4 Denaturation of the enzyme by the products.
5 Products are acting as competitive inhibitors.

A 1, 2 and 3
B 1, 4 and 5
C 2, 3 and 4
D 2, 4 and 5

5 A student examined the cells in the growing region (meristem) of an onion root and obtained the data below.

stage	**number of cells**
interphase	886
prophase	73
metaphase	16
anaphase	14
telophase	11

What percentage of cells contain chromosomes that appear as two chromatids?

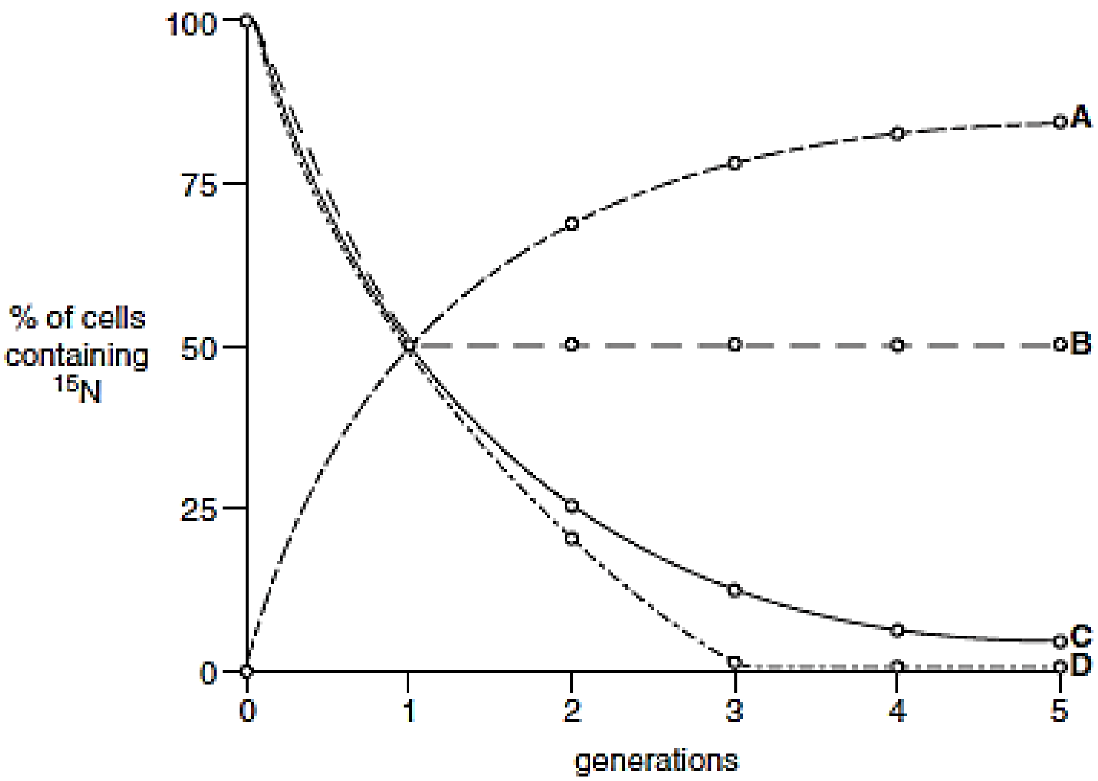
A 97.5%
B 95.9%
C 8.9%
D 7.3%

6 Which type of sugar and types of bonds are found in a DNA molecule?

	Type of sugar	type of bonds
A	non-reducing	hydrogen and ionic
B	non-reducing	hydrogen and peptide
C	reducing	covalent and hydrogen
D	reducing	hydrogen and peptide

7 Bacteria were cultured in a medium containing heavy nitrogen (^{15}N) until all the DNA was labelled. These bacteria were then grown in a medium containing only normal nitrogen (^{14}N) for five generations. The percentage of cells containing ^{15}N in each generation was estimated.

Which curve provides evidence that DNA replication is semi conservative? CCCCCCCCC



8 The following events occur during transcription.

1 Bonds break between complementary bases.
2 Bonds form between complementary bases.
3 Sugar-phosphate bonds form.
4 Free nucleotides pair with complementary nucleotides.

Before the mRNA leaves the nucleus, which events will have occurred twice?

A 1 and 2 only
B 1, 3 and 4 only
C 2, 3 and 4 only
D 1, 2, 3 and 4

9 A synthetic mRNA molecule is made by using only two types of nucleotide, containing adenine and cytosine. How many different codons could it contain?

A 2
B 4
C 8
D 16

10 A mutation of a gene coding for an ion pump in cell surface membranes results in the substitution of one amino acid, arginine, by another, histidine.

The DNA triplet codes for the two amino acids are shown.

Arginine	GCA	Histidine	GTA
	GCG		GTG
	GCT		
	GCC		
	TCT		
	TCC		

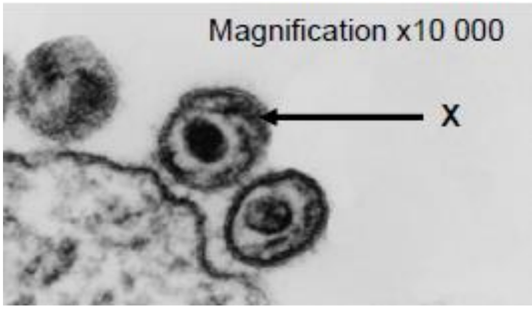
Which mutation has occurred in the DNA?

A Addition of an extra nucleotide
B Replacement of a nucleotide with a purine base by one with a different purine base
C Replacement of a nucleotide with a purine base by one with a pyrimidine base
D Replacement of a nucleotide with a pyrimidine base by one with a different pyrimidine base

11 The tobacco mosaic virus has RNA rather than DNA as its genetic material. In a hypothetical situation, RNA from a tobacco mosaic virus is mixed with proteins from a related DNA virus Z, resulting in a hybrid virus. If this hybrid virus were to infect a cell and reproduce, what would the resulting "offspring" viruses be like?

A Tobacco mosaic virus
B Virus Z
C A hybrid virus with tobacco mosaic virus RNA and virus Z protein
D A hybrid virus with virus Z DNA and tobacco mosaic virus protein

12 The diagram below shows an electron micrograph of HIV particles leaving a cell of an AIDS patient.



Which of following statements are false?

I. The size of HIV particle X is approximately 1.25 x 10^{12} nm.
II. The HIV particles recognise the host cell through the sialic-acid containing proteins or lipids on the membrane of the host cell.
III. Upon entry of the HIV into the host cell, a drop in pH will result in a conformational change in the HIV structure, consequently releasing the viral genome into the host cell.
IV. The viral DNA which enters the host cell's nucleus will be integrated into the genetic material of the host cell using the host cell's enzyme, integrase.

A II and IV only
B I and III only
C I, II and III only
D All of the above

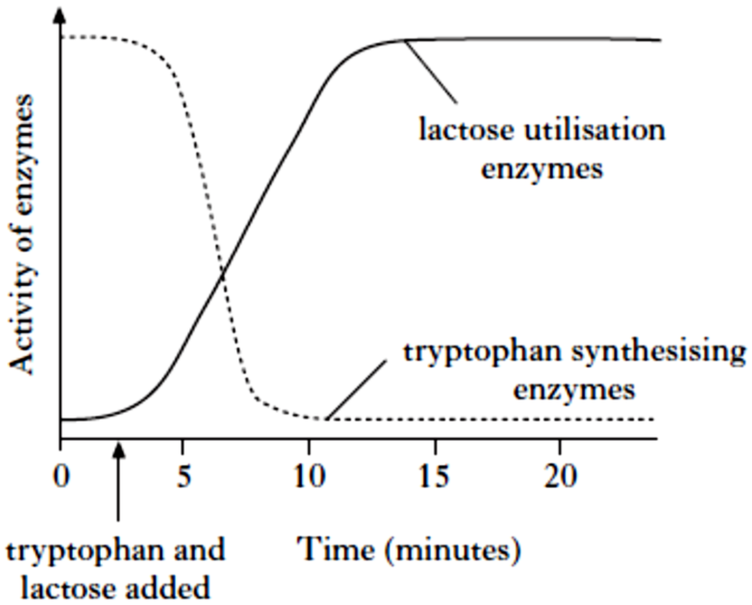
13 Which of the following processes would **never** contribute to genetic variation within a bacterial population?

A meiosis
B mutation
C conjugation
D transduction

14 A mutation that makes the regulatory gene of an inducible operon non-functional would result in

A continuous transcription of the operon's genes
B reduced transcription of the operon's genes
C accumulation of large quantities of a substrate for the catabolic pathway controlled by the operon
D Irreversible binding of the repressor to the promoter

15 The diagram below shows the changes in the activity of enzymes that synthesise tryptophan and utilise lactose in a cell after the addition of tryptophan and lactose.



What valid conclusion may be made from the graph?

A Addition of lactose acts as a negative enzyme modulator.
B Addition of tryptophan acts as a positive enzyme modulator.
C Enzyme induction is occurring in lactose utilisation enzymes.
D Enzyme induction is occurring in tryptophan synthesising enzymes.

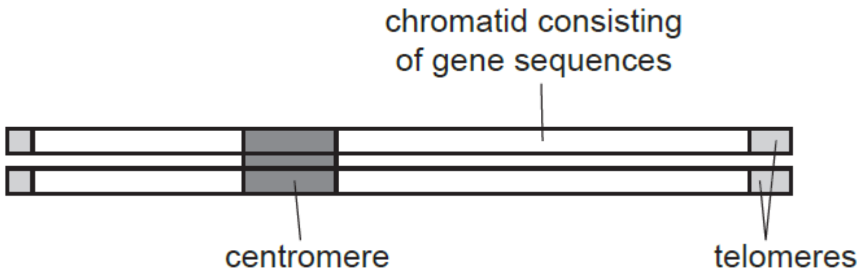
16 Which statement correctly describes the role of histone proteins?

A All eukaryotic genes are transcribed continuously because they are not packaged by histones.
B A DNA must be selectively released from its histone packaging before transcription can occur in bacteria.
C Histone package prokaryote chromatin into the nucleosomes that form the bulk of the chromosome.
D The organisation of DNA by histones in eukaryotes allows some gene control sequences to be thousands of base pairs away from the gene concerned.

17 In order to replicate, the ends of a eukaryotic chromosome contain a special sequence called a telomere. Human telomeres consist of repeating TTAGGG sequences which extend the ends of the chromosomal DNA.

When cells undergo mitotic division, some of these repeating sequences are lost. This results in the shortening of the telomeric DNA.

The diagram shows a eukaryotic chromosome.



What is a consequence of the loss of repeating DNA sequences from the telomeres?

A The cell will begin the synthesis of different proteins.
B The cell will begin to differentiate as a result of the altered DNA.
C The number of mitotic divisions the cell can make will be limited.
D The production of mRNA will be reduced.

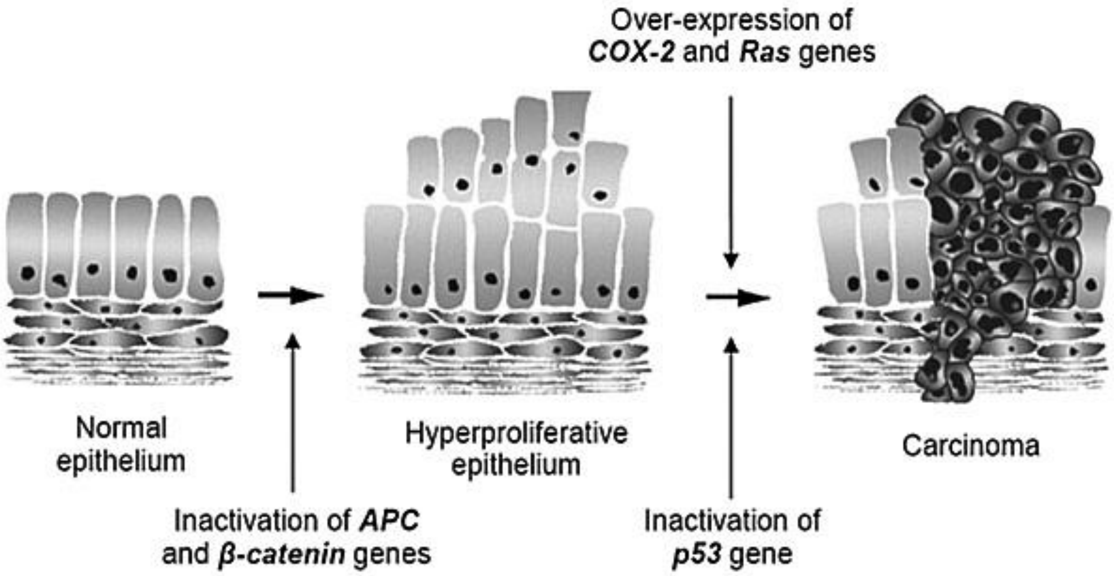
18 About 12,000 genes are expressed in both chick liver and oviduct. However an estimated additional 5,000 genes are expressed only in liver, while an additional 3,000 genes are expressed only in oviduct.

Which of the following could explain these observations?

I	The additional genes may have different methylation patterns in different tissues.
II	The concentrations of transcriptional enhancer elements for the additional genes vary in different tissues.
III	The number of genome copies is different in different somatic cells.
IV	A common set of genes are expressed for normal functions in liver and oviduct.

A **I** only
B **II** and **III** only
C **I** and **IV** only
D **II** and **IV** only

19 The diagram below illustrates the development of colorectal cancers.



Which of these statements can be inferred from this multistep model of carcinogenesis?

I. Cells whose *APC* and *β-catenin* genes are inactivated have lost density dependent inhibition.

II. *APC* and *β-catenin* genes are tumour suppressor genes.

III. High levels of *Ras* protein are produced only when both copies of *Ras* gene are mutated.

IV. Two copies of normal *p53* alleles must be present to inhibit cell division

V. Gain-of-function mutation in *COX-2* gene is a pre-requisite for the formation of carcinoma.

A **I**, **II** and **III**
B **II**, **III** and **IV**
C **I**, **II** and **V**
D **II, III** and **V**

20 A red-eyed, grey bodied fruit fly with the genotype **RRGG** was crossed with a brown-eyed, black-bodied fly with the genotype **rrgg**. The F_1 flies were allowed to interbreed to produce an F_2 generation (expecting a 9:3:3:1 ratio). A chi-squared test was carried out on the results obtained for the F_2 generation.

Part of the table of values for chi-squared is shown.

Degrees of freedom	$p \geq 0.5$	$p \geq 0.1$	$p \geq 0.05$	$p \geq 0.01$
1	0.46	2.71	3.84	6.64
2	1.39	4.60	5.99	9.21
3	2.37	6.25	7.82	11.34
4	3.36	7.78	9.49	13.28
5	4.35	9.24	11.07	15.09

The value of chi-squared in this investigation was 6.40.

What is the probability that chance explains the differences between the observed results and expected ratio?

	probability	*observed results fit expected ratio*
A	between 0.05 and 0.01	no
B	between 0.05 and 0.01	yes
C	between 0.05 and 0.1	no
D	between 0.05 and 0.1	yes

21 Two genes, Q and R, affect the size and pigmentation of the petals of flower.

Gene Q has two alleles, $\mathbf{Q^L}$ and $\mathbf{Q^A}$. The genotype of $\mathbf{Q^LQ^L}$ produces large petals, $\mathbf{Q^LQ^A}$ produces small petals and in $\mathbf{Q^AQ^A}$, petals are absent.

Gene R has two alleles, **R** produces a red pigment and is dominant over the allele **r** that produces no pigment.

Two plants, both heterozygous for both genes, are crossed.

How many phenotypes are expected in the next generation?

A 4
B 5
C 6
D 9

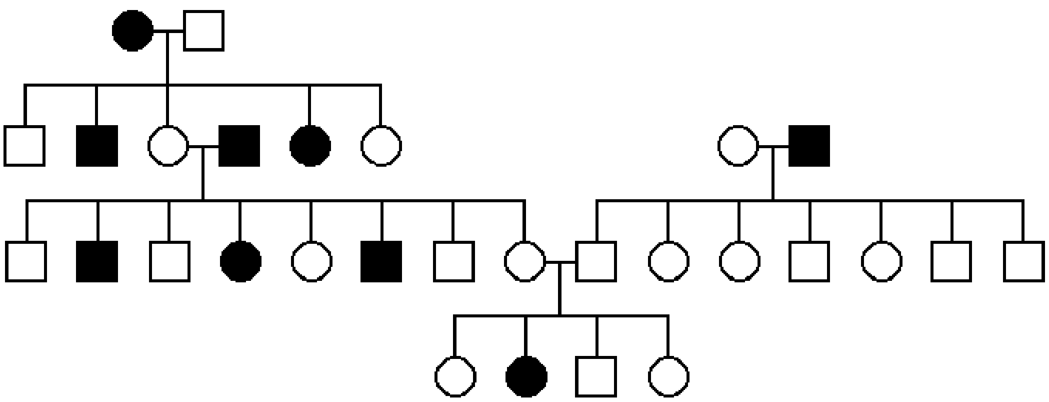
22 Possession of white or coloured feathers in poultry is controlled by two genes, P/p and Q/q.The phenotypes of offspring that are expected from mating two birds, each of which is heterozygous at both loci, are shown in the Punnett square.

gametes	PQ	Pq	pQ	pq
PQ	white feathers	white feathers	white feathers	white feathers
Pq	white feathers	white feathers	white feathers	white feathers
pQ	white feathers	white feathers	coloured feathers	coloured feathers
pq	white feathers	white feathers	coloured feathers	white feathers

What best explains the proportion of white to coloured feathers in the Punnett square?

A dominant epistasis in which the epistatic allele is P
B dominant epistasis in which the epistatic allele is Q
C recessive epistasis in which the epistastic allele is p
D recessive epistasis in which the epistatic allele is q

23 The inheritance of a genetic disease in a family is presented in a pedigree tree below.



What type of inheritance could this disease show?

I Autosomal dominant
II Autosomal recessive
III Sex-linked dominant
IV Sex-linked recessive

A I only
B II only
C I and III
D II and IV

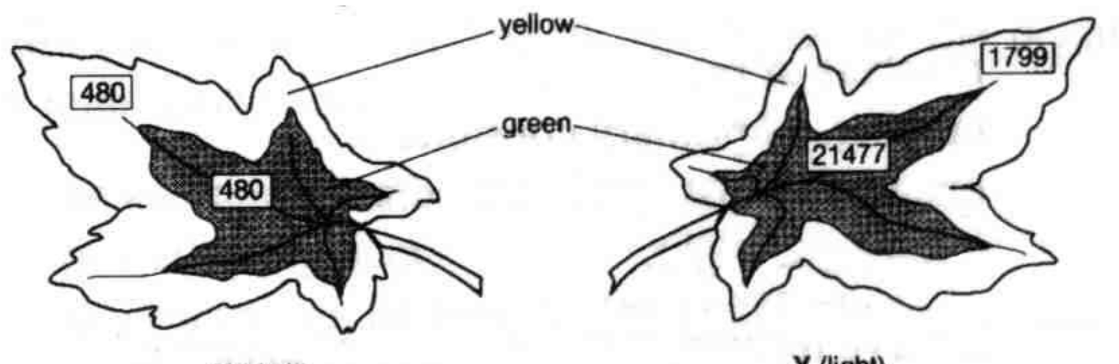
24 In plants adapted to cold conditions, their cell surface membranes change as the weather gets colder, allowing the plants to carry out exocytosis.

Which change occurs in their cell surface membranes?

A a decrease in the ratio of proteins to saturated phospholipids
B a decrease in the ratio of unsaturated phospholipids to saturated phospholipids
C an increase in the ratio of proteins to unsaturated phospholipids
D an increase in the ratio of unsaturated phospholipids to saturated phospholipids

25 Variegated leaves of a plant were supplied with radioactive carbon dioxide ($^{14}CO_2$) during an experiment. Leaf X was kept in the dark and leaf Y was kept in the light.

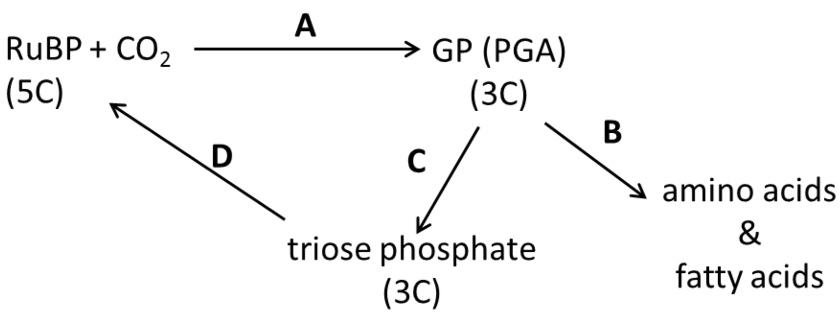
At the end of the experiment, the radioactivity in the leaves was measured. The results (in arbitrary units) are shown in the boxes in the diagrams.



What is the most likely explanation for the level of radioactivity found in the yellow zone of leaf Y?

A Photosynthesis occurs but no storage of starch occurs in this zone.
B Photosynthesis proceeds slowly in the absence of chlorophylls a and b.
C Products of photosynthesis are transported into the yellow zone.
D Radioactive carbon dioxide diffuses into the leaf and accumulates there.

26 The diagram shows stages in the light independent reaction of photosynthesis. CCCCCCC



At which stage is most of the reduced NADP oxidised?

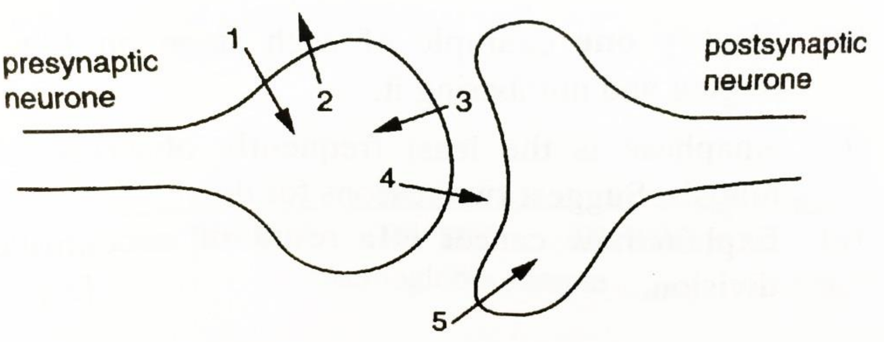
27 ATP can be formed in the following ways.

1 Oxidative phosphorylation in the electron transport system
2 Glycolysis

In the complete oxidation of one molecule of glucose, approximately what percentage of ATP comes from 1?

A 25%
B 50%
C 80%
D 90%

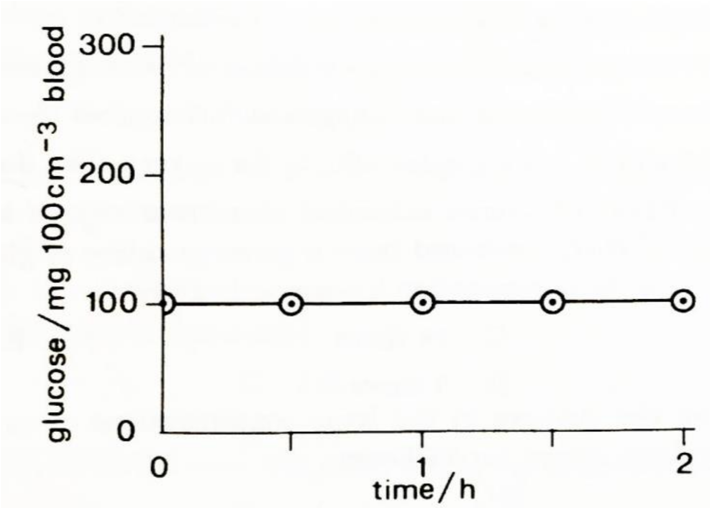
28 The diagram shows the sequence of events occurring as an action potential arrives at a synapse. The numbered arrows represent the movement of substances across the membranes.



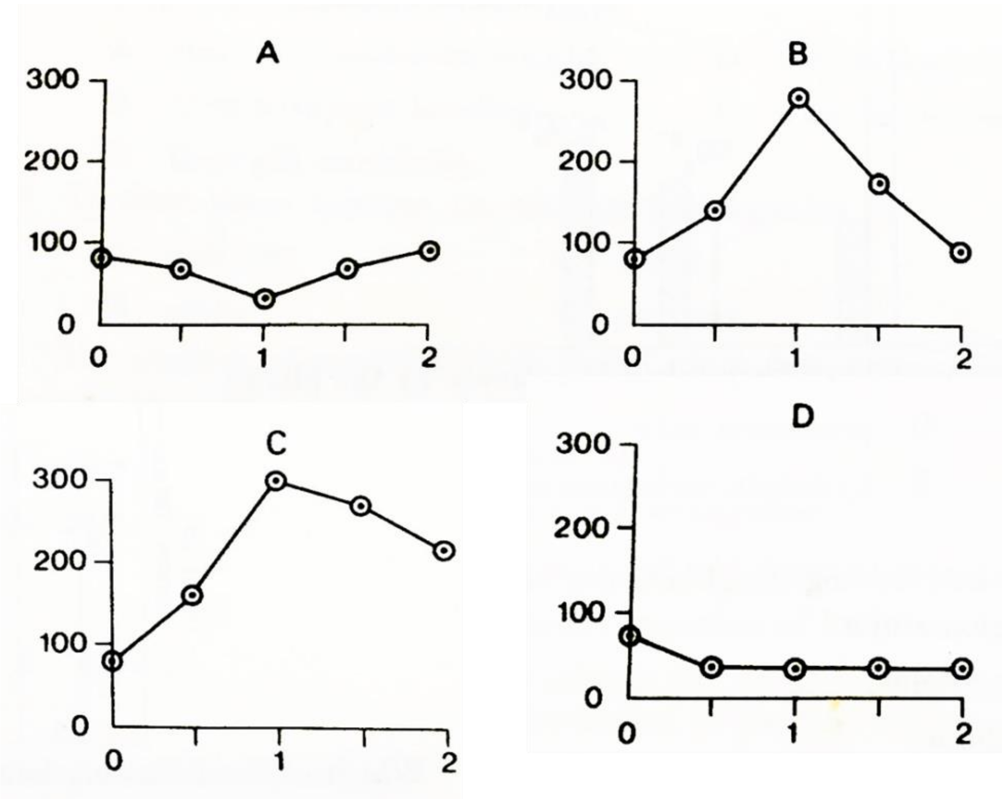
What are the substances moving across the membranes?

	1	2	3	4	5
A	K^+	Na^+	acetylcholine	Ca^{2+}	K^+
B	K^+	Na^+	K^+	Ca^{2+}	acetylcholine
C	Na^+	K^+	Ca^{2+}	acetylcholine	Na^+
D	Na^+	K^+	Na^+	acetylcholine	Ca^{2+}

29 The graph below illustrates the changes in blood sugar concentration after a healthy man has drunk a glucose solution. CCCCCCC



Which one of the following graphs would you apply to a diabetic man in similar circumstances?



30 Red light can only penetrate about 10m through water, but blue and green wavelengths can penetrate more than twice that distance.

Pigments which can absorb light of one colour and use the energy to emit light of another colour are called 'fluorescent'. Many reef corals living at depths between 10 and 20m fluoresce red.

Some fish living between 10 and 20m also have pigments which fluoresce red. These pigments often surround the eyes and form patches on the body and fins, in patterns unique to each species.

Which of these suggested reasons could explain the selective advantage of fluorescence for these fish?

1 Fluorescent patches provide camouflage.
2 Recognition of their own species avoids interbreeding with other species.
3 The fluorescent patches do not attract predators more than 10m away.

A 1,2 and 3
B 1 and 2 only
C 1 and 3 only
D 2 and 3 only

31 Deer mice are small, ground-dwelling rodents. They normally have dark coats.

Some paler coated mice have been observed in an area where new sand hills were formed between 8000 and 15 000 years ago. The sand hills are sparsely covered with scrubby plants. Studies have shown that the change in coat colour was due to a mutation in a gene, causing a single amino acid deletion from the protein for the coat colour pigment. This mutation seems to be spreading through the population.

Which statement most fully explains how the evolution of this species at this site has occurred?

A Better camouflage increases the chance of survival by reducing predation.
B Deer mice which live longer usually leave more offspring since they have more reproductive opportunities .
C The occurrence of the mutation provided new variation on which natural selection can act.
D The paler colour gives better camouflage against the background of the new sand hills.

32 The diagram shows part of the aligned DNA sequences for the same gene in six species of aquatic mammals.

Fin whale	TAAACCCCAATAGTCACAAAACAAGACTATTCGCCAGAGTACTACTAGCAAC
Humpback whale	TAAACCCTAATAGTCACAAAACAAGACTATTCGCCAGAGTACTACTAGCAAC
Sperm whale	TAAACCCAGGTAGTCATAAAACAAGACTATTCGCCAGAGTACTACTAGCAAC
Beaked whale	TAAACCTAAATAGTCTCAAAACAAGACTATTCGCCAGAGTACTACTAGCAAT
Dolphin	TAAACTTAAATAATCCCAAAACAAGATTATTCGCCAGAGTACTATCGGCAAC
Porpoise	TAAACCTAAATAGTCCTAAAACAAGACTATTCGCCAGAGTACTATCGGCAAC

Which is a correct assumption when using this information as evidence for evolutionary relationships?

A Differences and similarities in DNA sequences reflect evolutionary relationships.
B DNA sequences in different genes from the same six species will suggest different evolutionary relationships.
C Point mutations in DNA sequences are not acted on by natural selection.
D Mutations that do not change amino acid sequences in proteins are important for natural selection.

33 Many types of evidence, including homology, can provide evidence in support of Darwin's theory of natural selection.

Which statement does **not** provide support?

A The allele for sickle cell haemoglobin that gives resistance to malaria is more frequent in malarial areas.
B The distribution of the variants of the A blood group antigen reflects human migration patterns.
C The homozygous condition of the sex-linked allele for a non-functional blood clotting protein is rare.
D The molecular structure of ATP is almost identical in all eukaryotes.

34 BamHI is a restriction enzyme produced by *Bacillus amyloliquefaciens.* It recognises the specific nucleotide sequence GGATCC. *B. amyloliquefaciens* is susceptible to attack by viruses known as bacteriophages.

Which statement describes the natural function of BamHI?

A BamHI enzymatic modification of nucleotides occurs in all the GGATCC nucleotide sequences of *B. amyloliquefaciens,* to protect the DNA from degradation by bacteriophage enzymes and prevent its nucleotides being used for viral nucleic acid synthesis.

B Infectivity rate is lowered when *B. amyloliquefaciens* is subjected to invasion by bacteriophages that have double-stranded DNA as their genetic material, since BamHI can recognise and cleave specific viral nucleotide sequences.

C Restriction fragments produced as a result of BamHI binding to and cleaving DNA at all sites within the bacterial cell that have a GGATCC nucleotide sequence, prevent integration of bacteriophage nucleic acid into the bacterial genome.

D Transcription and translation of the *B. amyloliquefaciens* gene encoding the BamHI enzyme is followed by secretion of the enzyme into the external environment, where it has antibacterial action against non-related bacterial species.

35 The restriction enzyme *Tai1* , cuts DNA as shown.

```
----ACGT----   →   ----ACGT                ----
----TGCA----       ----                TGCA----
```

Which of the plasmids below can be cut using restriction enzyme *Tai* 1, in order to allow DNA to be inserted into the circular plasmid without loss of any part of the plasmid? Aaaaa

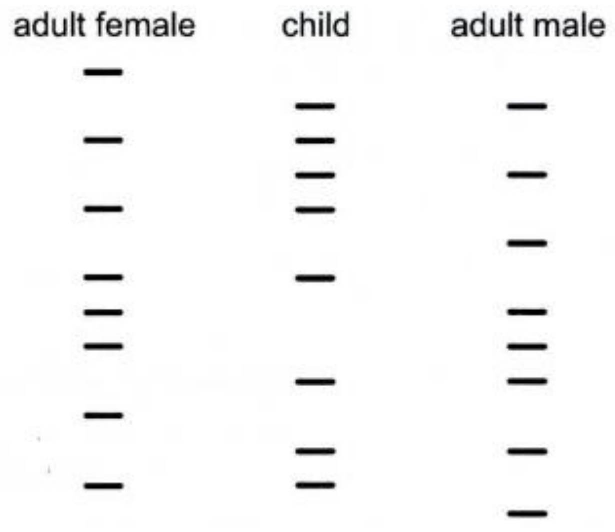
```
A  ----TAACGTAC----CTCAAGCT----
   ----AATGCATG----GAGTTCGA----

B  ----TTAACGTA----CACGTGGT----
   ----AATTGCAT----GTGCACCA----

C  ----ACGTACGT----TCCACGTA----
   ----TGCATGCA----AGGTGCAT----

D  ----ACCTAGGT----CCACCTGA----
   ----TGGATCCA----GGTGGACT----
```

key

---- other nucleotides

The rest of the plasmid does not contain this restriction site.

36 The diagram shows the results of a DNA analysis, carried out by electrophoresis. The DNA of three individuals was profiled. The diagram shows the results of a DNA analysis, carried out by electrophoresis. The DNA of three individuals was profiled.



What is the **most** likely conclusion from these results?

A The child cannot be related to either the male or female.
B The female cannot be the child's mother.
C The male cannot be the child's father.
D The male and female could be the child's parents.

37 Which statements are goals of the human genome project (HGP)?

1 To map the positions of all the genes on all the human chromosomes. This would help to identify the location of many genes that can cause disease.

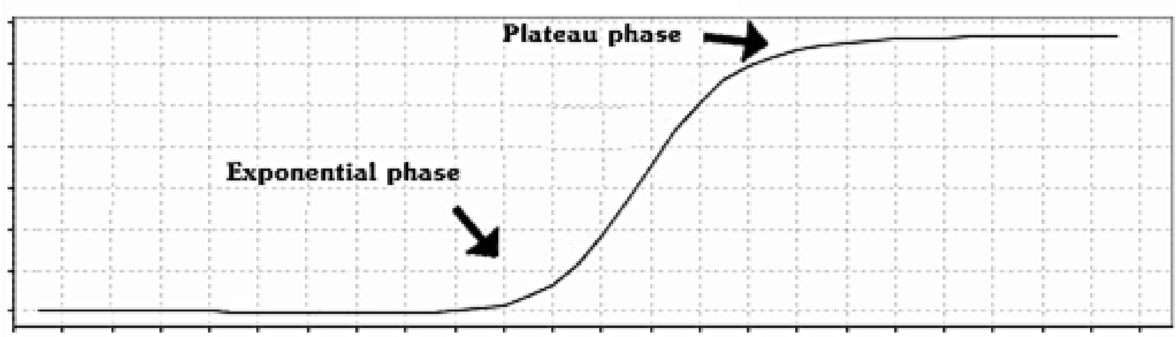
2 To obtain the sequence of all the DNA nucleotides on all the human chromosomes. This would enable particular sequences to be patented so that advancements could be made in gene therapy.

3 To allow the various centres sequencing the DNA nucleotides on all the human chromosomes to share results on the internet. This would ensure that the data were publicly available.

4 To develop techniques and technological tools for use in the sequencing project. This would lead to faster sequencing of sections of the human genome and improvements in mapping.

A 1, 2 and 3 only
B 1, 2 and 4 only
C 1, 3 and 4 only
D 2, 3 and 4 only

38 During the process of polymerase chain reaction (PCR), the amount of DNA synthesised can be traced using fluorescent probes and the measurements are shown in the following plot. The process initially goes through an exponential phase, followed by a plateau phase eventually.



Which of the following statements is **true**?

A During the exponential phase, the number of DNA molecules synthesized after 15 cycles is 15^2.

B During the exponential phase, the temperature is always maintained at the optimum temperature of 72°C hence there is rapid amplification.

C During the plateau phase, the reaction mixture is being depleted of ribonucleotides.

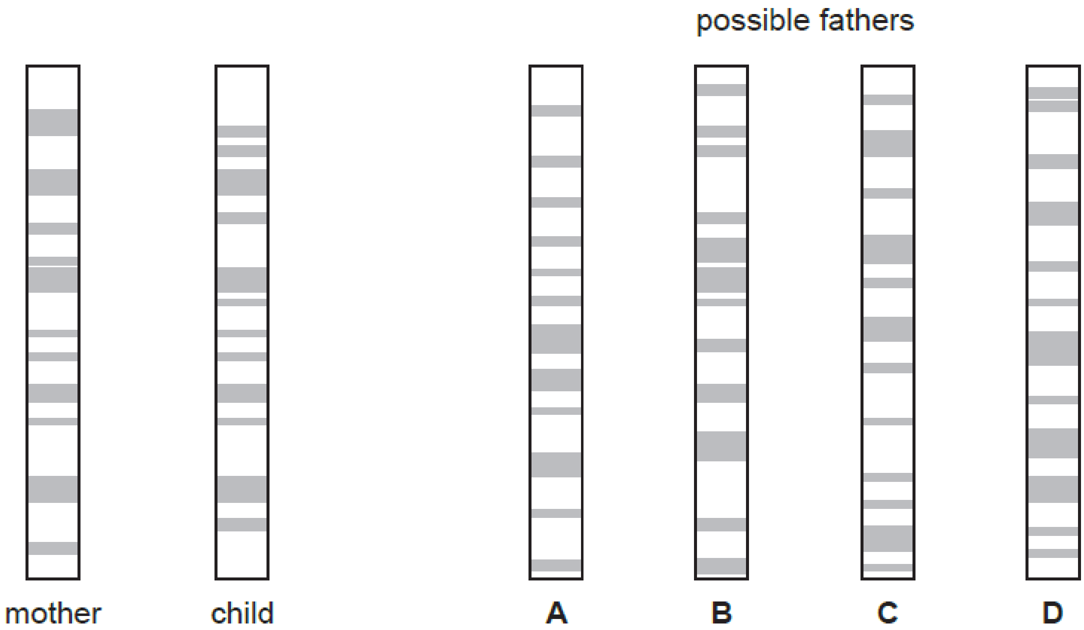
D During the plateau phase, *Taq* polymerase may be denatured.

39 Genetic profiling can be used to determine the paternity of a child.

DNA from the mother and the child is cut into fragments separated by electrophoresis and made visual using a stain.

The diagram shows the genetic profiles of a mother and child, and four possible fathers.

Who is the father? BBBBBBB



40 Marker genes are often inserted into genetically engineered crop plant cells, along with desired genes. Bacterial antibiotic resistance genes are sometimes used as marker genes. These may include short DNA repeats to make them unstable so that they are quite quickly eliminated by the genetically engineered crop plant cells.

Which is not a reason why elimination of such marker genes is favoured?

A It is theoretically possible for the antibiotic resistance marker gene in human food to pass to bacteria in the human gut.

B It is difficult to carry out repeated transformations using the same antibiotic.

C The antibiotics may affect the growth and differentiation of the fields of crop plants.

D There are a few such antibiotic resistance marker genes available.